Task 1: Daily Water Consumption

Create a line chart showing liters of water consumed during a week.

```
days = ["Mon","Tue","Wed","Thu","Fri","Sat","Sun"]
water = [2,3,2,4,3,5,4]
```

Requirements

- Create line chart
- Add title
- Add X label
- Add Y label
- Save chart as:

```
water_consumption.png
```

Task 2: Mobile Usage Hours

```
days = ["Mon","Tue","Wed","Thu","Fri"]
hours = [2,4,3,5,4]
```

Requirements

- Use line chart
- Show markers
- Enable grid

Task 3: Student Attendance Trend

```
weeks = ["Week1","Week2","Week3","Week4"]
attendance = [70,75,85,90]
```

Questions:

1. Which week has highest attendance?
2. Which week has lowest attendance?

Plot the chart.

Level 2: Bar Charts

Task 4: Subject Marks

```
subjects = ["Python","SQL","Statistics","English"]
marks = [88,92,75,80]
```

Create a bar chart.

Requirements

- Proper title
- X label
- Y label
- Save chart

Task 5: Monthly Expenses

```
expenses = {
    "Food": 5000,
    "Travel": 2500,
    "Shopping": 3000,
    "Bills": 2000
}
```

Create a bar chart.
Question:
Which category has highest expense?

Task 6: Department Employee Count

```
departments = ["HR","IT","Finance","Marketing"]
employees = [15,40,20,25]
```

Create bar chart.

Level 3: Scatter Plots

Task 7: Study Hours vs Marks

```
study_hours = [1,2,3,4,5,6]
marks = [40,50,60,70,80,90]
```

Create scatter plot.
Question:
What relationship do you observe?

Task 8: Attendance vs Marks

```
attendance = [60,70,80,90,95]
marks = [55,65,75,85,95]
```

Create scatter plot.

Task 9: Exercise vs Calories Burned

```
exercise_hours = [1,2,3,4,5]
calories = [150,250,350,450,550]
```

Create scatter plot.

Level 4: Data Analysis + Visualization

Task 10: Student Result Analysis

Create DataFrame:

```
import pandas as pd

students = {
    "Name":["Asha","Ravi","Kiran","John","Divya"],
    "Marks":[88,76,84,79,95]
}

df = pd.DataFrame(students)
```

Questions

1. Find average marks.
2. Find highest marks.
3. Find lowest marks.
4. Plot student marks.

Task 11: Employee Salary Dashboard

```
employees = {
    "Employee":["A","B","C","D","E"],
    "Salary":[35000,42000,50000,38000,60000]
}
```

Tasks

- Create DataFrame
- Find average salary
- Plot salary chart

Task 12: Product Sales Dashboard

```
sales = {
    "Product":["Laptop","Mouse","Keyboard","Monitor"],
    "Sales":[50,150,100,75]
}
```

Tasks

- Convert to DataFrame
- Plot bar chart
- Find best-selling product

Level 5: GroupBy Practice

Task 13: Branch-wise Average Marks

```
data = {
    "Student":["A","B","C","D","E","F"],
    "Branch":["CSE","ECE","CSE","EEE","ECE","CSE"],
    "Marks":[80,70,90,85,75,95]
}
```

Tasks

1. Create DataFrame
2. Calculate average marks branch-wise

```
df.groupby("Branch")["Marks"].mean()
```

1. Plot bar chart

Task 14: Department-wise Salary Analysis

```
data = {
    "Employee":["A","B","C","D","E"],
    "Department":["IT","HR","IT","Finance","HR"],
    "Salary":[50000,40000,60000,55000,45000]
}
```

Tasks

- Group by department
- Calculate average salary
- Plot chart